

Noughts and Crosses

2 · 1 minute · nothing but a pencil

Rule two vertical lines and two horizontal lines to make nine cells. The first player writes a cross in any cell, the second an nought, and so on alternately. The winner is the first to hold three cells in a straight line — across, down, or corner to corner. If all nine cells are filled with no line, the game is a draw, which is what it will be every time once both of you can play. Play it as a race instead: keep a running score across twenty games and count only the wins, so a draw costs the player who let it happen. Swap who goes first each game, because the first player has the whole of the advantage. Nine cells is the standard, but the rules work unchanged on any square grid, and on a larger one the draw stops being automatic.

Dots and Boxes

2 to 4 · 20 minutes · nothing but a pencil

Draw a square array of dots — six by six gives twenty-five boxes and a proper game. Players alternate, and a move draws one short line joining two horizontally or vertically neighbouring dots. Diagonals are not allowed. If your line completes the fourth side of a box, you claim it: write your initial inside, and **move again**. Completing several boxes in one turn earns several extra moves, so a single line can be worth a dozen boxes. When every line has been drawn, the player holding more boxes wins. With three or four players the rules are unchanged and the alliances are informal. If you want the original ending, Lucas's own rule was that the game stops the moment one player holds more than half the boxes.

Sprouts

2 · 10 minutes · nothing but a pencil

Start by drawing three or four spots anywhere on blank paper. A move does two things at once: draw a curve joining one spot to another — or from a spot back to itself — and then place a new spot somewhere along the curve you have just drawn. Three restrictions govern everything. The curve may bend as much as you like but may not cross itself, cross any other curve, or pass through any spot. No spot may ever have more than **three** curve-ends touching it; a loop from a spot to itself uses two of its three, and the new spot you add is born with two already used. Play alternately. The player who cannot move loses. A game from n spots lasts at least $2n$ moves and at most $3n - 1$, so it always ends.

Sim

2 · 5 minutes · nothing but a pencil

Draw six dots arranged in a circle, well spaced. There are fifteen possible lines joining pairs of them, and every one of the fifteen will eventually be drawn. Players alternate, each drawing one line per turn, and each player's lines must be distinguishable — use solid for one player and dashed for the other, since a pencil gives you no colours. The rule is a prohibition, not an objective: **you lose if you complete a triangle all three of whose sides are your own** and whose corners are three of the six dots. Triangles made of mixed lines are harmless, and so are the crossing points where lines meet in the middle of the figure — only the six original dots count as corners. The game can never be drawn.

Sea Battle

2 · 20 minutes · squared paper and a pencil

Each player draws two ten-by-ten grids, lettered A to J across and numbered 1 to 10 down: one is your own sea, the other your record of your opponent's. Secretly mark your fleet on your own grid — one ship of four squares, two of three, three of two, four of one — placed horizontally or vertically, never diagonally, and never touching one another, not even at a corner. Players then alternate calling a single square, say **D7**. Your opponent must answer **miss**, **hit**, or **sunk** when the last square of a ship goes. Record every answer on your second grid: a dot for a miss, a cross for a hit. The winner is the first to sink the whole enemy fleet. Agree before you start whether a hit earns another shot; both conventions are common.

Bulls and Cows

2 · 10 minutes · nothing but a pencil

Each player secretly writes down a four-digit number whose digits are **all different**. Players then take turns guessing the other's number, and each guess must also have four different digits. After each guess the setter reports two counts and nothing else: the number of **bulls**, meaning digits that are correct and in the right position, and the number of **cows**, meaning digits that appear in the number but in a different position. So if the secret is 4271 and the guess is 1234, the answer is one bull and two cows — the bull is the 2, the cows are the 4 and the 1. The setter never says which digits. The first player to be told **four bulls** has won.

Jotto

2 · 20 minutes · nothing but a pencil

Each player secretly writes a five-letter word with no repeated letter. Players alternate guessing, and each guess must also be a real five-letter word with no repeated letter. The setter answers with ****one number only****: how many letters the guess has in common with the secret word, regardless of position. No positional information is ever given — that is the difference between this and its digit cousin. So if the secret is HORSE and you guess SHORE, the answer is five, and you still do not know the word. Keep the alphabet written along the top of your page and cross letters off as you eliminate them. First to name the other's word wins. Agree in advance whether proper nouns are allowed; the usual answer is no.

Ghost

2 or more · 10 minutes · nothing but a pencil

Players take turns adding one letter to a growing fragment, writing it down so everyone can see. Every fragment must be the beginning of some real word, but ****you must not complete a word****: the player whose letter finishes a word of four letters or more loses the round. Instead of adding a letter, you may challenge the previous player to name a word beginning with the current fragment. If they can, the challenger loses the round; if they cannot, they do. A lost round earns you a letter of the word GHOST, and a player who spells the whole word is out. The last player left wins. Bluffing is legal and, played well, is most of the game.

Categories

2 or more · 15 minutes · nothing but a pencil

Everyone rules the same grid: five or six columns headed with categories the group agrees on — a country, an animal, a boy's name, something you eat, a word for money — and one row per round. Someone picks a letter at random. Everyone then writes, as fast as they can, one entry in each column beginning with that letter. The first player to fill every column calls ****stop****, and everybody must put their pencil down. Score together: an answer nobody else gave is worth two points, an answer shared with someone else is worth one, a blank or a duplicate of your own earlier answer is worth nothing. Play ten rounds with ten different letters. The Guggenheim version uses a keyword down the side and one letter per row.

Doublets

2 or more · 15 minutes · nothing but a pencil

Two words of the same length are set — the ***doublet***. Each player must get from the first to the second by changing ****one letter at a time****, where every intermediate step is itself a real word and the letters stay in their own places, never swapping around. The intermediate words are the ***links***, and the whole run is the ***chain***. Carroll's own example turns HEAD into TAIL by way of heal, teal, tell and tall. Give everyone three minutes and score by the length of the chain: the shortest valid chain wins, and a player with no chain at all scores nothing. Good doublets to set: PIG to STY, FOUR to FIVE, WHEAT to BREAD, HATE to LOVE.

Nim

2 · 5 minutes · nothing but a pencil

Draw several rows of strokes — the classic layout is four rows of one, three, five and seven. On your turn choose ****one**** row and cross out as many strokes from it as you like, from one stroke up to the whole row, but never from more than one row in a single turn. Players alternate. The player who crosses out the last stroke on the board ****wins****. In the *misère* version, agreed before you start, that player loses instead, and it is a different game with a different rule. Nim is worth playing on paper rather than with matches for one reason: the crossed-out strokes stay visible, so the history of the game is on the page and you can see exactly where it was lost.

Bachet's Hundred

2 · 3 minutes · nothing but a pencil

Two players take turns saying a number from ****one to ten****, and each number is added to the running total, which is written down after every turn. The player who brings the total to exactly ****one hundred**** wins. That is the whole game, and it takes twenty seconds to teach and thirty years to forget. Everything about it can be adjusted: the target may be any number you like, and so may the cap on what a player may say. Bachet's own examples are a target of 120 with a cap of ten, and a target of 100 with a cap of eight. His fourth example, a target of 100 with a cap of nine, is the interesting one, because it is the only one of the four the first player loses.